

PEAS – Performance Measure, Environment, Actuators, Sensors

PEAS is a method used to define an intelligent agent by clearly specifying four things:

1. Performance Measure:

- It tells how we judge if the agent is doing a good job.
- Example: For a vacuum cleaner agent, performance measure may be amount of dirt cleaned, time taken, electricity saved.

2. Environment:

- The world in which the agent works.
- Example: Rooms, floor type, obstacles, dirt.

3. Actuators:

- The parts through which the agent takes action.
- Example: Vacuum brush, wheels, suction motor.

4. Sensors:

- The parts through which the agent receives input.
- Example: Dirt sensor, bump sensor, location sensor.

Simple Example (Self-Driving Car):

- Performance Measure: Safety, speed, comfort.
- Environment: Roads, traffic, signals.
- Actuators: Steering, brakes, accelerator.
- Sensors: Cameras, GPS, radar.

PEAS helps in designing AI agents by clearly defining what they must achieve, what world they work in, and how they sense and act.